



A predictive model for flow distribution in dual-zone packed bed reactors

Pablo Díaz^{ID}, Consuelo Reinao, Iván Cornejo^{ID}*

Departamento de Ingeniería Química y Ambiental, Universidad Técnica Federico Santa María, Vicuña Mackenna 3939, Santiago, 8940897, Chile

ARTICLE INFO

Keywords:

Dual-zone packed bed
Flow distribution
Flow control
CFD
Ergun equation

ABSTRACT

This paper proposes a new predictive model for flow distribution in dual-zone packed bed reactors, which consist of two concentric regions, an inner core and an outer ring, each containing pellets with the same shape but different sizes. This configuration aims to manipulate flow paths for further improving reactor performance. The proposed model, derived from the Ergun equation, captures how the combination of different pellet diameters, relative core sizes and inlet conditions influence the flow distribution between the two regions of the bed. The model's predictions are compared against reactor-scale CFD data across a wide range of Reynolds numbers and bed geometry combinations, showing strong agreement for all the tested conditions. According to the results, different terms of the Ergun equation dominate in different regions of the bed, influencing the local flow distribution and pressure drop. This insight highlights the importance of considering these varying effects when optimizing the packed bed reactor design. This study provides a theoretical framework to better understand and control flow distribution in dual-zone packed beds, with potential applications in catalytic chemical processes. The proposed model offers a useful engineering tool for novel packed reactor configurations.

1. Introduction

In the design of catalytic packed bed reactors, controlling flow distribution is essential for enhancing performance and preventing catalyst degradation, particularly in exothermic systems prone to hot spots. Hot spots are especially critical, for example, in emerging renewable synthesis processes, such as the conversion of green hydrogen and captured CO₂ into methanol [1,2], the synthesis of dimethyl ether (DME) [2], biomass pyrolysis [3,4], and other value-added chemical syntheses. Flow maldistribution leads to localized regions where reactants convert at higher rates, creating zones of elevated temperature. This issue is further exacerbated by interphase resistances between the fluid and solid phases, causing maldistribution even in uniformly packed beds. The result is a rapid increase in temperature, particularly in adiabatic reactors with limited heat dissipation [5,6]. In some cases, maldistribution causes regions of low flow, where reactants accumulate and undergo faster reactions, generating excessive heat. As these reactants leave the low flow regions, they mix with cooler components from other parts of the bed, leading to temperature drops. Such uncontrolled variations can result from physical obstructions or irregularities in catalyst packing such as fines posing significant challenges to reactor stability [5,6].

Achieving a uniform temperature profile in catalytic beds is essential not only for maximizing reaction efficiency but also for prolonging

catalyst lifespan. Hot spots within the reactor can deactivate catalysts, reduce selectivity, and promote by-product formation, collectively diminishing process efficiency [7–9]. Current strategies to mitigate these issues include diluting the reactant concentration, lowering the inlet temperature, using smaller catalyst particles, and employing multiple adiabatic beds with inter-stage cooling. However, these approaches often come at the cost of reduced productivity, higher pressure drops, or increased capital and operating expenses [10,11]. For instance, while reducing catalyst particle size can mitigate hot spots, it significantly increases the pressure drop across the bed, raising operational costs. Similarly, increasing the bed's thermal conductivity by mixing catalysts with inert materials helps diffuse temperature gradients but also leads to higher pressure drops, complicating operations [10,11]. Although these solutions effectively stabilize reactor performance, they highlight the inherent trade-offs between safe reactor operation and optimal efficiency. As a result, achieving better flow distribution without compromising productivity remains a core challenge in packed bed reactor design.

As alternatives to traditional packed bed reactors, various support configurations have been explored. One promising approach involves combining pellets with open-cell foams, particularly for non-adiabatic catalytic processes that require high heat transfer rates [12–14]. Packing pellets within open-cell foams enhances thermal conductivity while

* Corresponding author.

E-mail address: ivan.cornejo@usm.cl (I. Cornejo).

<https://doi.org/10.1016/j.cej.2025.162962>

preserving sufficient catalyst volume [13,14]. Research indicates that the packing efficiency and pressure drop in these systems depend heavily on the ratio between the foam window size and the pellet diameter. Larger pellets tend to improve packing efficiency, while smaller pellets increase porosity and promote local channeling. Notably, despite the foam structure, the pressure drop in packed foam systems is often comparable to or even lower than that in traditional packed beds with the same pellet size, owing to the increased porosity [12]. Nonetheless, several challenges remain in the application of foams in catalytic reactors. While foams improve heat transfer, they can sometimes compromise mass transfer, particularly in systems where high reactant flow rates are necessary [15–17]. Another challenge is the mechanical stability of the foam structure, a high porosity may cause deformation or collapse over time, especially under high pressure and temperature conditions [18–20]. Addressing these challenges requires further optimization of foam materials and configurations to ensure consistent performance and longevity in industrial applications [20].

Monolith supports have garnered significant attention in recent years. They consist of continuous solid structures with numerous channels running in parallel. This design not only reduces pressure drop but also improves flow uniformity and temperature distribution across the channels [21]. Among these designs, non-homogeneous core/ring monolith substrates featuring different cell densities in the core and periphery have shown great potential for enhancing flow control. By adjusting geometrical variables, these configurations may further potentially optimize temperature profiles and reactor performance [22, 23].

Mixing pellet sizes has been shown to significantly influence flow behavior in industrial processes. When particles vary in size, predicting the pressure drop becomes more challenging due to complex fluid-particle interactions, particularly when the size ratio between particles is large. The Ergun equation is traditionally employed to estimate pressure drop in packed beds but shows limitations when applied to systems with polydisperse particle size distributions [24,25]. Despite these complexities, fluid dynamics research often assumes uniform particle sizes, as it remains a valid approximation for many practical applications.

In the analysis of packed bed reactors, pressure drop is a key factor influenced by flow distribution and plays a critical role in reducing operating costs. The Ergun equation [26] is the most widely used model for predicting pressure drop in packed beds [27], though some studies highlight its limitations. Specifically, it has been shown to overpredict pressure drop when the particle Reynolds number (Re_p) exceeds 300 [27]. To improve its accuracy, several modifications have been proposed. Some researchers have adjusted the equation's constants to fit specific experimental conditions, as seen in the work of Macdonald et al. [28], KTA [29], and Ozahi et al. [30]. Others have incorporated additional terms to account for porosity, such as in the models by Rose [31] and Gibilaro et al. [32]. Wall effects also become significant in smaller reactors, where the ratio of tube diameter to particle size is reduced, prompting models that address these effects, such as those by Montillet [33] and Harrison et al. [34]. Although no consensus or general formula has been established across these models, the Ergun equation remains the standard in predictive studies involving packed beds [1,4,35], and it is also adopted for the purposes of this work. A more detailed description and analysis of these models, along with others, can be found in [27].

The Ergun equation [26] consists of two terms. The first is linear and corresponds to the Kozeny–Carman equation, which applies to low Reynolds numbers ($Re_p < 10$). The second is quadratic, known as the Burke-Plummer equation, and is relevant at high Reynolds numbers ($Re_p > 1000$) [36]. These terms capture different physical phenomena. The linear term accounts for pressure losses due to viscous forces, while the quadratic one describes losses because of the inertial forces. Together, they model the cumulative pressure drop resulting from

solid-fluid interactions across a wide range of flow regimes [26]. The complete equation is expressed as follows:

$$\frac{\Delta p}{L} = \frac{150\mu(1-\varepsilon)^2}{\Phi^2 D_p^2 \varepsilon^3} u + \frac{1.75\rho(1-\varepsilon)}{\Phi D_p \varepsilon^3} u^2 \quad (1)$$

Recent studies have proposed alternative packed-bed reactor configurations, such as using square channels or operating with a small tube-to-particle diameter ratio, to better understand transport phenomena and mitigate thermal hotspots in catalytic processes [37,38]. These approaches, however, often require refinements to Ergun-like correlations, particularly by modifying inertial terms or accounting for wall effects in non-standard packings. Together, these cases underscore that the Ergun framework is still evolving through continuous refinements, supporting that pressure-drop modeling in complex packed-bed geometries remains an interesting topic for the community.

Selecting an appropriate pressure drop model is essential for developing a flow distribution model in dual-zone reactors, such as the one proposed by Reinao et al. [23]. This approach relies on two key principles: (i) mass balance, where the inlet flow is divided between the core and ring regions, and (ii) backpressure equilibrium, where the flow adjusts such that the pressure drop across both regions becomes balanced. In this configuration, a region with lower backpressure receives more flow, increasing its pressure drop until equilibrium is achieved. However, the application of these concepts to packed bed reactors remains largely unexplored. Driven by the need for efficient and reliable catalytic processes, this study aims to address the gap in understanding flow distribution in non-homogeneous packed bed reactors.

This investigation hypothesizes that a strategic mixture of pellet sizes in the core and peripheral (ring) regions, as depicted in Fig. 1, may effectively control flow distribution and, in turn, regulate the temperature profile. This configuration targets preventing central hot spots in the bed, where temperatures tend to be highest due to the distance from external cooling. While traditional hot spots due to local irregularities may still occur within the core or ring regions, this approach manipulates flow distribution to reduce the likelihood of overheating in the bed's center, enhancing reactor performance by tailoring flow resistance in each region. However, designing such reactors through preliminary calculations requires accurate predictions of the residence time of reactants in each zone, a challenge that relies on the absence of engineering models capable of predicting flow distribution in dual-zone packed beds. Despite the maturity of Ergun-like models, surprisingly, no explicitly available models address *core-ring* concentric configurations such as the one covered in this study. The lack of a dedicated framework for dual-zone packed beds presents a gap for systematic exploration of such configurations, particularly in terms of explicitly quantifying the influence of local parameters such as particle size on, for example, flow distribution. This study addresses this gap by developing the first predictive model tailored for these systems, providing a relevant tool for preliminary reactor design. The proposed model aims to enable more accurate flow predictions and support the optimization of catalytic reactor performance.

In the presented context, the impact of varying pellet sizes between the core and ring regions of the bed is investigated using CFD. This configuration aims to create disparities in apparent permeability and inertial resistance, enabling controlled flow distribution. A reactor-scale model of the core/ring bed is employed to obtain flow distribution data and compare it with predictions from a theoretical model. First, the computational model is described along with a parametric study evaluating the effects of pellet size, inlet Reynolds number, and core size. Then, a theoretical model, based on the backpressure equilibrium and Ergun equation, is developed to capture the balance in pressure drop between regions with different particle sizes, facilitating a stable distribution of flow. The results from both the computational and theoretical approaches are then compared to assess the accuracy and applicability of the proposed model.

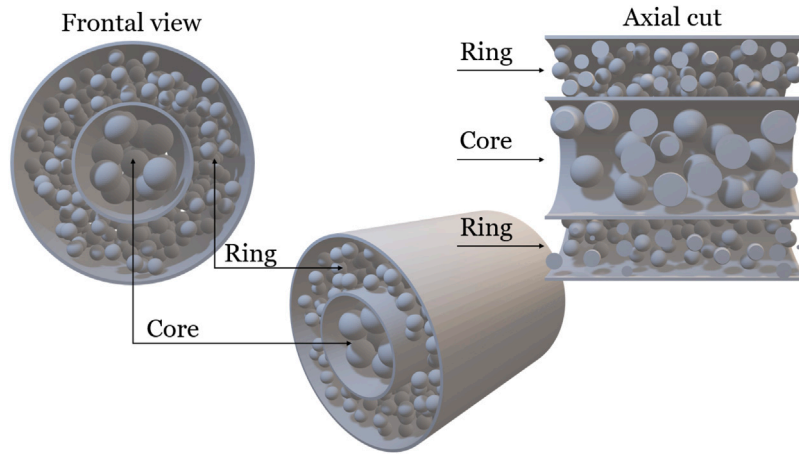


Fig. 1. Dual-zone packed bed with different particle sizes in core and ring regions.

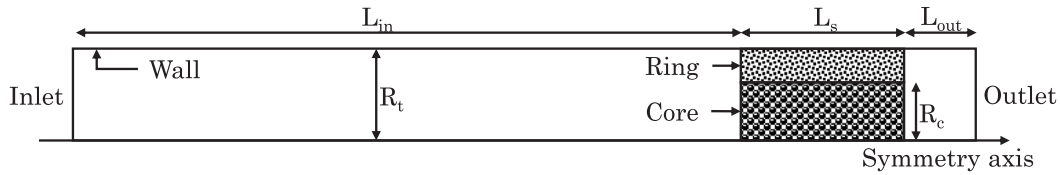


Fig. 2. Schematic of the computational domain.

2. Computational model

This section presents the development and implementation of the computational model used to analyze flow distribution and pressure drop in dual-zone packed beds.

2.1. Description of the domain

The computational domain was set to represent the dual-zone packed bed configuration from Fig. 1, allowing for variations in flow resistance between the core and ring regions to evaluate their impact on flow distribution. A schematic of the axisymmetric section used to generate the domain is shown in Fig. 2. The domain is divided into three sections: an inlet space before the pellet bed, an outlet part after the bed, and a middle section containing the packed bed. The lengths of these sections were $L_{in} = 1000$ mm, $L_{out} = 20$ mm, and $L_s = 150$ mm, respectively. The overall system radius (R_t) was 35 mm. The packed bed consisted of spherical pellets with a uniform void fraction of 0.4, which is, in general terms, independent of the particle diameter [39,40]. The core and ring regions feature different apparent permeability and inertial resistance due to variations in particle size.

The bed is modeled using a continuum approach, where the substrate is treated as a homogeneous porous medium with a representative apparent permeability and porosity. This approach simplifies the simulation by significantly reducing computational cost, making it suitable for systematic research. It leverages the axial symmetry of the domain to focus on macroscopic flow behaviors rather than particle-scale interactions. While a DEM-CFD coupled model could provide more detailed insights into individual particle interactions [41], it goes beyond the scope of the current investigation, taking into account that the primary objective of the study is to analyze the overall flow distribution, for which the continuum approach is well-established and validated in packed bed reactor analysis.

2.2. Flow model

The flow regime through the domain was assumed to be turbulent and was modeled using a Reynolds-Average Navier–Stokes (RANS) model. For steady state and absence of gravitational effects, the continuity and momentum conservation equations were [42]:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (2)$$

$$\frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_i}{\partial x_i} \delta_{ij} \right] - \frac{\partial \tau_{ij}}{\partial x_j} + S_{u_i} \quad (3)$$

The closure was provided by the Boussinesq “eddy viscosity” approximation and the SST $k - \omega$ model [42–44]. Different turbulence models for packed bed reactors have been studied in the literature, but this is still an open research topic. Despite this, SST RANS-based models have shown good representation of real phenomena and low computational cost [45,46].

The Reynolds stress tensor was [42–44]:

$$-\tau_{ij} = -u'_i u'_j = \mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_i}{\partial x_i} \delta_{ij} \quad (4)$$

The turbulence viscosity was [42–44]:

$$\mu_t = \rho \frac{k}{\omega} \frac{1}{\max \left(\frac{1}{\alpha^*}, \frac{SF_2}{a_1 \omega} \right)} \quad (5)$$

The transport equations for turbulence kinetic energy (k) and specific turbulence dissipation rate (ω) from SST model were [42–44]:

$$\frac{\partial(\rho k u_i)}{\partial x_j} = \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k - Y_k \quad (6)$$

where

$$G_k = \min(\mu_t S^2, 10\rho\beta^* k\omega) \quad (7)$$

$$Y_k = \rho\beta^* k\omega \quad (8)$$

$$\frac{\partial(\rho\omega u_i)}{\partial x_j} = \left[\left(\mu + \frac{\mu_t}{\sigma_\omega} \right) \frac{\partial\omega}{\partial x_j} \right] + G_\omega - Y_\omega + D_\omega \quad (9)$$

and

$$G_\omega = \alpha_\omega \alpha^* \rho \mu_t S^2 \quad (10)$$

$$Y_\omega = \rho \beta^* \omega^2 \quad (11)$$

$$D_\omega = 2(1 - F_1) \rho \frac{1}{\omega \sigma_{\omega_2}} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (12)$$

Further details about constants and functions included in the SST model can be found in [43,44]. The term S_{u_i} in Eq. (3) was used to account for the extra head loss just in the porous medium due to the flow passing through the packed bed and was modeled by the Darcy–Forchheimer’s law [47], as shown in Eq. (13).

$$S_{u_i} = - \left(\frac{\mu}{\alpha_i} u_i + C_i \frac{1}{2} \rho u_i |u_i| \right) \quad (13)$$

The axial and radial components of S_{u_i} are equal in each zone, indicating that the medium is isotropic. This isotropy allows the flow to be treated similarly in both axial and radial directions, as in a flow-through packed bed. The apparent permeability and inertial resistance for each region are determined using Eqs. (14) and (15). In these equations, the subscript n denotes the core (c) and ring (r) regions, respectively.

$$\alpha_{axial_n} = \alpha_{radial_n} = \frac{\Phi^2 D_p^2 \epsilon^3}{150(1 - \epsilon)^2} \quad (14)$$

$$C_{axial_n} = C_{radial_n} = \frac{3.5(1 - \epsilon)}{\Phi D_p \epsilon^3} \quad (15)$$

The problem was implemented using the commercial CFD code ANSYS Fluent 2023R2 [42], which employs the finite volume method. Atmospheric air was used as the working fluid, with a constant density of 1.225 kg/m³ and a constant viscosity of 1.7894×10^{-5} Pa·s. The top boundary was defined as a no-slip wall for momentum, while the bottom boundary, representing the tube’s centerline, was set as a symmetry axis. The inlet velocity had a flat profile and varied in velocity magnitude across different runs, while the outlet boundary condition was set to a prescribed pressure of 1 bar.

For interpolating, a second-order upwind scheme was used for momentum, and the pressure–velocity coupling was solved using a pseudo-transient, fully coupled approach. The convergence criteria for every run was having scaled residuals below 10^{-4} , along with a stable pressure drop value.

2.3. CFD model validation and grid independence

This section is intended to validate the CFD framework, providing a spatially distributed reference for comparison with the proposed one-dimensional theoretical model. The grid independence study was performed by comparing the pressure drop across different meshes, using a particle diameter of 1 mm and an inlet Reynolds number (Re_{in}) of 10000. The grid, fully composed of rectangular elements, was progressively refined, maintaining a constant aspect ratio of 1:2 between the vertical and horizontal dimensions of the control volumes, which aligns with the flow direction. The results are presented in Fig. 3. The difference in pressure drop between the second-to-last mesh (329 940 control volumes) and the finest mesh (517 725 control volumes) was only 0.04%. Since this difference is negligible, the mesh with 329 940 control volumes was selected for further simulations.

Given the absence of models for flow distribution in non-homogeneous fixed beds in the literature, the developed computational model was validated by comparing its pressure drop predictions with those obtained from Ergun’s equation (Eq. (1)) for flow through a packed bed, as described in Section 1.

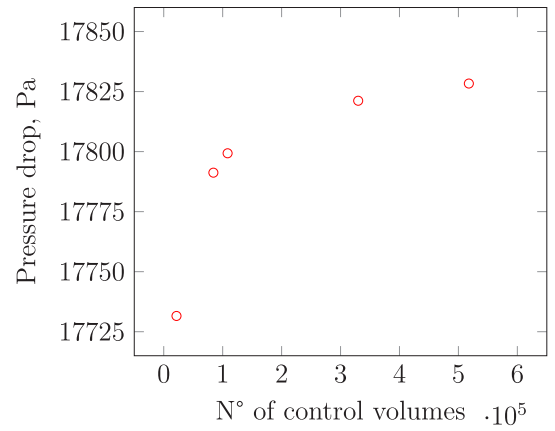


Fig. 3. Pressure drop vs. number of control volumes for grid independence study.

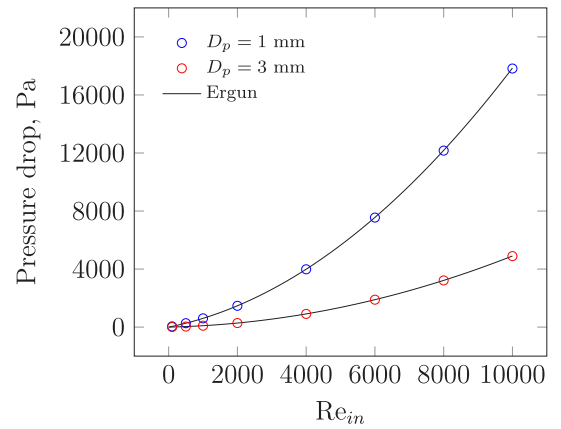


Fig. 4. Comparison between computational model's pressure drop and that calculated from Ergun equation.

The pressure drop data from the computational model were calculated by measuring the pressure difference between the inlet and outlet of the packed bed. To maintain consistency with the momentum balance, mass-averaged total pressure values were used across these sections. Using total pressure is essential for accurately determining losses between regions with different velocity profiles [23]. The pressure at a specific axial position was determined using the following expression:

$$p|_x = \frac{\int_A p u dA}{\int_A u dA} \quad (16)$$

Simulations were conducted for two particle diameters, 1 mm and 3 mm, over a range of Re_{in} from 100 to 10000, as shown in Fig. 4. The computational model's predictions showed good agreement with the Ergun equation for both particle sizes.

The results in Fig. 4 highlight that for both 1 mm and 3 mm particle diameters, the computational model captures the expected non-linear relationship between the pressure drop and Re_{in} . As Re_{in} increases, the pressure drop rises more sharply, reflecting the transition from viscous-dominated to inertial-dominated flow, consistent with the behavior predicted by the Ergun equation. This agreement across a wide range of conditions demonstrates the robustness of the computational model for predicting pressure losses in dual-zone packed beds.

Table 1

List of computational experiments. For all of them: $D_{p,r} = 2$ mm. For a core size of 100%, the value of $D_{p,c}/D_{p,r}$ is only indicated to report $D_{p,c}$, since there is no ring region in these cases.

N	Re _{in}	$D_{p,c}/D_{p,r}$	Core Size, %	N	Re _{in}	$D_{p,c}/D_{p,r}$	Core Size, %
1	100	0.50	25	31	100	0.50	75
2	100	0.75	25	32	100	0.75	75
3	100	1.00	25	33	100	1.00	75
4	100	1.25	25	34	100	1.25	75
5	100	1.50	25	35	100	1.50	75
6	1000	0.50	25	36	1000	0.50	75
7	1000	0.75	25	37	1000	0.75	75
8	1000	1.00	25	38	1000	1.00	75
9	1000	1.25	25	39	1000	1.25	75
10	1000	1.50	25	40	1000	1.50	75
11	10000	0.50	25	41	10000	0.50	75
12	10000	0.75	25	42	10000	0.75	75
13	10000	1.00	25	43	10000	1.00	75
14	10000	1.25	25	44	10000	1.25	75
15	10000	1.50	25	45	10000	1.75	75
16	100	0.50	50	46	100	0.50	100
17	100	0.75	50	47	100	0.75	100
18	100	1.00	50	48	100	1.00	100
19	100	1.25	50	49	100	1.25	100
20	100	1.50	50	50	100	1.50	100
21	1000	0.50	50	51	1000	0.50	100
22	1000	0.75	50	52	1000	0.75	100
23	1000	1.00	50	53	1000	1.00	100
24	1000	1.25	50	54	1000	1.25	100
25	1000	1.50	50	55	1000	1.50	100
26	10000	0.25	50	56	10000	0.50	100
27	10000	0.50	50	57	10000	0.75	100
28	10000	1.00	50	58	10000	1.00	100
29	10000	1.25	50	59	10000	1.25	100
30	10000	1.50	50	60	10000	1.50	100

3. Results and discussion

This section summarizes the main results obtained. As described by the Ergun equation, the principal variables affecting pressure drop, and consequently flow distribution, inside a packed bed are the inlet (superficial) velocity, represented by Re_{in}, and particle diameter combination. These variables were selected for the parametric study, along with core size, in alignment with previous investigations on dual-cell density monolith systems [23]. Three values were chosen for Re_{in}: 100, 1000, and 10000, which correspond to particle Reynolds numbers (Re_p) ranging from 1 to 500, covering a broad spectrum of values commonly reported in the literature [27,33,48]. The particle diameter in the ring region was held constant at 2 mm, while the core particle size was varied to obtain relative particle diameters ($D_{p,c}/D_{p,r}$) ranging from 0.50 to 1.50, with increments of 0.25. Core size, defined as the percentage of the bed cross-sectional area assigned to the core region, was varied from 25% to 100% in increments of 25%. It is important to note that when the core size is 0% and 100%, the system corresponds to a homogeneous packed bed with the particle diameter corresponding to $D_{p,r}$ and $D_{p,c}$ respectively. By combining all possible parameter values mentioned, a total of 60 simulations were performed, as listed in Table 1.

The core velocity was obtained from CFD simulations, and the percentage of flow passing through the core region was calculated using Eq. (17). This percentage represents the fraction of the total inlet flow directed through the core for an incompressible fluid, assuming constant density:

$$f_c = \frac{u_c A_c}{u_{in} A_T} 100\% \quad (17)$$

3.1. Exploratory analysis

The first result presented is an exploratory, qualitative analysis to corroborate some model assumptions, such as the backpressure equilibrium between the core and ring of the bed.

Table 2

Pressure drop and particle Reynolds number for the conditions in Table 1.

N	Δp , Pa	Re _{p,c}	Re _{p,r}	N	Δp , Pa	Re _{p,c}	Re _{p,r}
1	15.44	0.46	3.50	31	28.96	0.86	6.25
2	13.99	1.39	3.19	32	18.46	1.82	4.14
3	12.45	2.86	—	33	12.45	2.86	—
4	11.00	4.76	2.54	34	8.87	3.90	2.07
5	9.72	6.99	2.26	35	6.62	4.94	1.56
6	232.43	6.27	33.91	36	419.40	10.54	51.02
7	205.06	15.98	30.99	37	260.80	19.37	36.82
8	183.50	28.57	—	38	183.48	28.57	—
9	166.88	43.00	26.63	39	139.35	37.93	23.24
10	153.78	58.71	25.05	40	111.37	47.37	19.54
11	9320.16	97.46	315.98	41	14 106.17	124.75	394.38
12	8380.46	185.72	298.41	42	9915.24	204.03	326.72
13	7732.26	285.71	—	43	7731.67	285.71	—
14	7238.46	394.75	275.68	44	6376.09	368.99	257.28
15	6840.38	511.22	267.35	45	5445.52	453.43	235.99
16	20.23	0.61	4.50	46	48.49	1.43	—
17	15.94	1.58	3.61	47	21.84	2.14	—
18	12.45	2.86	—	48	12.45	2.86	—
19	9.83	4.29	2.28	49	8.07	3.57	—
20	7.90	5.80	1.85	50	5.68	4.29	—
21	305.71	8.01	41.13	51	602.86	14.29	—
22	230.57	17.56	33.73	52	297.05	21.43	—
23	183.49	28.57	—	53	183.46	28.57	—
24	152.24	40.36	24.86	54	127.91	35.71	—
25	130.20	52.61	22.07	55	96.10	42.86	—
26	11 365.28	109.87	351.69	56	17 821.17	142.86	—
27	9105.72	194.54	312.05	57	10 833.75	214.29	—
28	7732.02	285.71	—	58	7731.23	285.71	—
29	6787.67	381.51	266.22	59	5996.63	357.14	—
30	6086.70	480.85	250.86	60	4892.67	428.57	—

Fig. 5 shows the pressure distribution for the section of the domain close to the substrate. It includes cases with different core sizes, Reynolds numbers, and relative pellet diameters. Notably, within the bed region, the pressure remains largely uniform along the radial direction, maintaining similar values in both the core and ring zones. This observation confirms that backpressure equilibrium is achieved under the evaluated conditions. Additionally, the pressure drop increases when the core size is smaller, as the extent of the more permeable region is reduced. Examining the influence of the Reynolds number, a comparison between Fig. 5(a) and (c) shows that a higher Reynolds number results in a more significant pressure drop. Furthermore, the comparison of Fig. 5(a) and (d) indicates that a relative particle diameter lower than 1.00 leads to an increase in pressure drop.

Fig. 6 presents the velocity field within the same aforementioned section of the domain. In the bed region, the velocity remains relatively consistent and with different magnitudes for the core and ring zones. When the core size is reduced, the flow distribution variations become more pronounced, leading to an increase in velocity through the core. Additionally, a lower Reynolds number results in a reduced inlet velocity, which subsequently decreases the overall flow rate through the bed, affecting the distribution and velocity magnitudes in both regions. It is also noted that as the relative particle diameter increases, the flow preferentially moves through the more permeable region as the resistance contrast between the core and ring intensifies.

To complement previous results, Table 2 summarizes the pressure drop (Δp), together with the core ($Re_{p,c} = \rho u_c D_{p,c}/\mu$) and ring ($Re_{p,r} = \rho u_r D_{p,r}/\mu$) particle Reynolds numbers for the runs in Table 1. It is important to note that for a given inlet Reynolds number, the particle Reynolds numbers in the core and ring regions can vary significantly depending on the core size and particle diameter ratio. For instance, in case 30 of Table 2, the particle Reynolds number in the core is nearly twice that in the ring. This variation should be considered when analyzing potential flow regime transitions in physical systems.

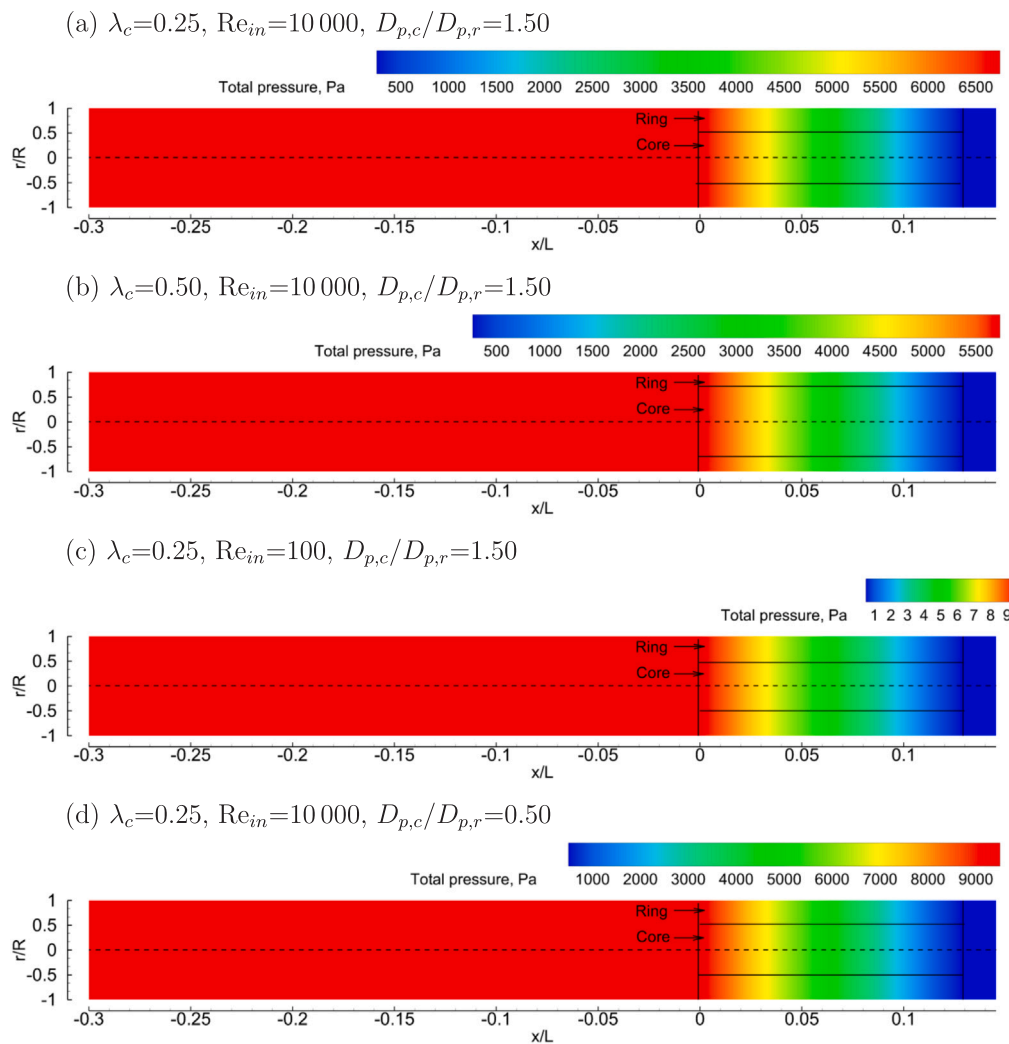


Fig. 5. Pressure drop distribution for different core sizes, tube Reynolds number and relative pellet diameter. The x-axis is scaled as $x/(L_{in}+L_s+L_{out})$.

3.2. Effect of the core size

For the effect of core size, values ranging from 25% to 100% were implemented, increasing 25% each time. Since a core size of 100% is equivalent to 0%, as both represent homogeneous packed beds, these cases were included in the results of this section. The ratio $D_{p,c}/D_{p,r}$ was kept fixed at 0.5, with the ring particle diameter set at 2 mm. Fig. 7 shows the percentage of flow passing through the core (f_c) as a function of core size for various Re_{in} values. A positive correlation between f_c and core size is observed, indicating that larger core regions allow a greater fraction of the flow to pass through.

Reynolds number (Re_{in}) significantly affects the flow distribution between the core and ring regions. At $Re_{in} = 100$, the value of f_c is lower across all core sizes compared to higher Reynolds numbers. As Re_{in} increases, the percentage of flow passing through the core also rises, reflecting a stronger dependence of flow distribution on inertial forces at higher velocities. The sensitivity of f_c to Re_{in} depends on the size of the core. For smaller cores (25% and 50%), the differences in f_c between different Re_{in} values are relatively minor, suggesting that at small core sizes, viscous forces play a more dominant role. As the core size increases to 75% and 100%, these differences become more pronounced, indicating that larger cores are more sensitive to variations in Re_{in} . Notably, at 100% core size (homogeneous bed), all Re_{in} values

converge since, in such a condition, there is an absence of a dual-zone configuration.

This analysis suggests that larger cores, excluding the homogeneous case, exhibit greater sensitivity to changes in flow conditions, as inertial effects may become more dominant. However, the particle size ratio ($D_{p,c}/D_{p,r}$) also plays a critical role in flow behavior, which will be addressed in the following sections.

3.3. Effect of the tube Reynolds number

The effect of tube Reynolds number (Re_{in}) on flow distribution was examined by fixing the core size at 25%. The particle size ratio ($D_{p,c}/D_{p,r}$) was varied from 0.50 to 1.50 in increments of 0.25, with the ring particle diameter set at 2 mm. Re_{in} values of 100, 1000, and 10000 were considered.

As shown in Fig. 8, the percentage of flow passing through the core (f_c) varies across different particle size ratios ($D_{p,c}/D_{p,r}$) and tube Reynolds numbers (Re_{in}). At lower Re_{in} values, f_c is more sensitive to particle size ratio, with larger ratios (e.g., $D_{p,c}/D_{p,r} = 1.50$) allowing more flow through the core. This behavior reflects the dominance of viscous forces at lower Reynolds numbers. However, as Re_{in} increases, the differences in f_c across particle size ratios become less pronounced, indicating that inertial forces dominate at higher velocities, reducing the influence of particle diameter.

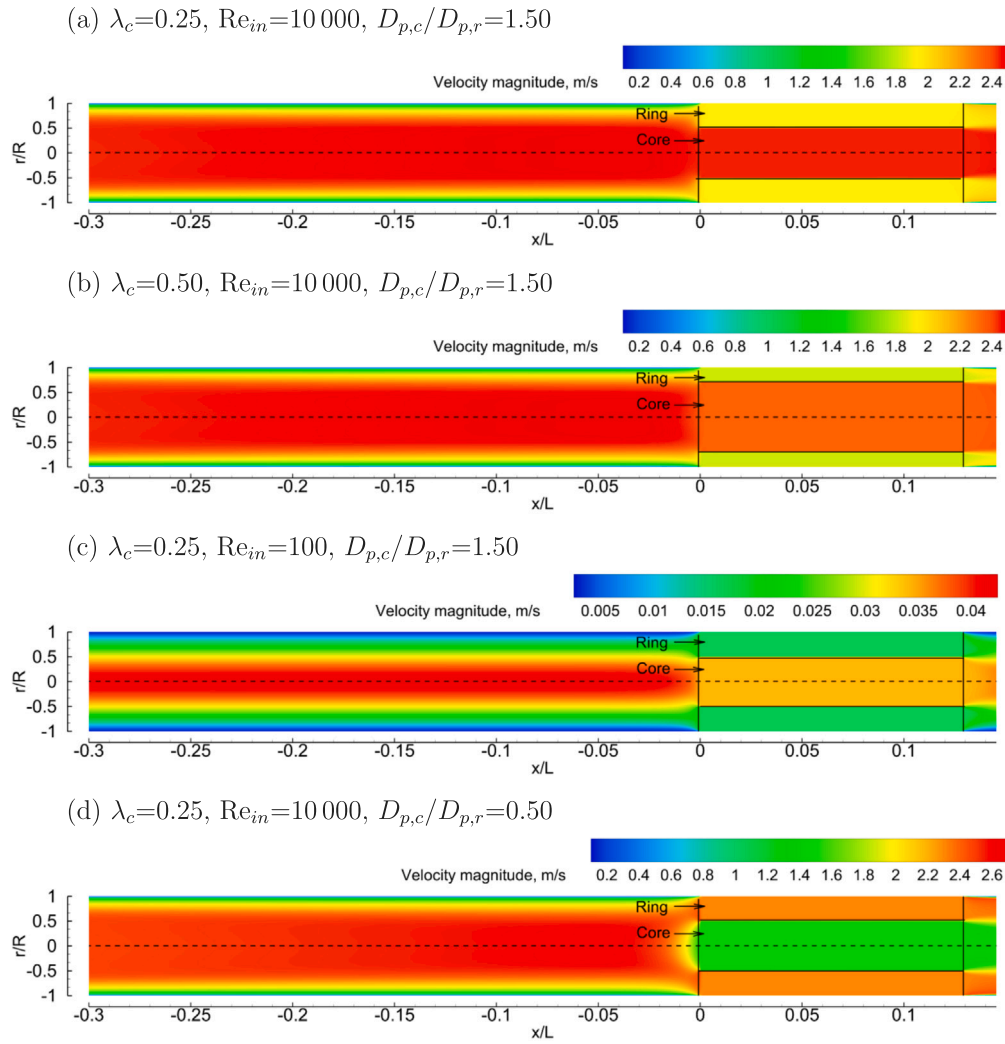


Fig. 6. Velocity distribution for different core sizes, tube Reynolds number and relative pellet diameter. The x-axis is scaled as $x/(L_{in}+L_s+L_{out})$.

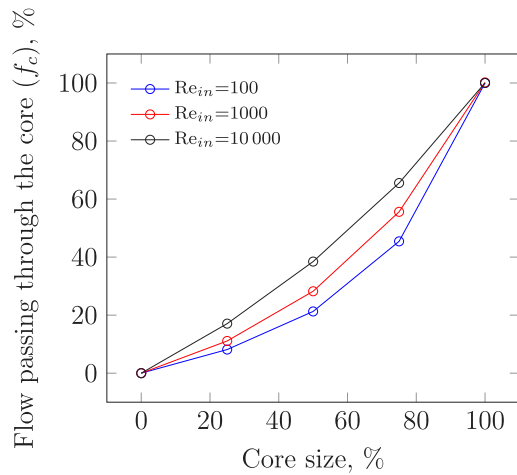


Fig. 7. Flow percentage through the core (f_c) as a function of core size for different Re_{in} values. $D_{p,c}/D_{p,r}=0.5$.

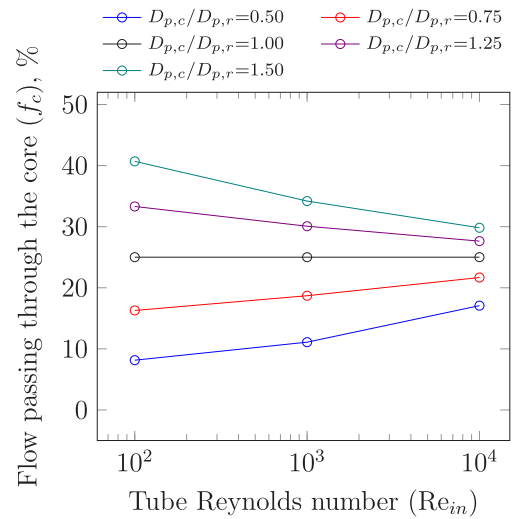


Fig. 8. Flow percent passing through the core in function of tube Reynolds number in logarithmic scale with $\lambda_c = 25\%$.

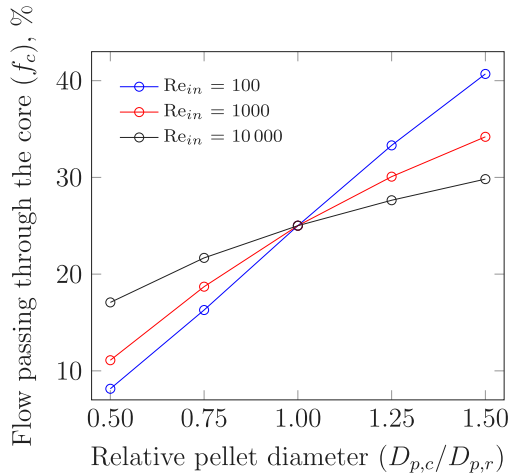


Fig. 9. Flow percent passing through the core in function of relative particle size with $\lambda_c = 25\%$.

The trends in f_c are non-linear, with larger particle size ratios sustaining higher core flows even at elevated Re_{in} . As Re_{in} increases toward 10 000, the flow percentages across different particle size ratios tend to converge. This convergence suggests that beyond a certain Reynolds number, flow distribution becomes less sensitive to particle size ratio, as inertial forces dominate.

3.4. Effect of the relative particle diameter

The last variable studied was the relative particle diameter ($D_{p,c}/D_{p,r}$). For this analysis, combinations of previously used parameters were tested under two conditions. Firstly, a core size fixed at 25%, and secondly, a Re_{in} fixed at 100. The results are presented in Figs. 9 and 10, respectively.

As shown in Fig. 9, $D_{p,c}/D_{p,r}$ has a significant effect on the percentage of flow through the core (f_c). For particle size ratios of 0.50 and 0.75, f_c is lower across all Reynolds numbers, although it increases as Re_{in} rises. At a ratio of 1.0, f_c equals the core size, representing the homogeneous packed bed case where the core and ring have equal particle diameters. For larger ratios (1.25 and 1.50), f_c increases for all Reynolds numbers, although the growth in flow becomes less significant as Re_{in} increases. This behavior reflects the decreasing influence of particle diameter ratio on flow distribution at higher Reynolds numbers.

The trends in Fig. 10 further demonstrate how increasing the particle size ratio enhances core flow. This is explained by differences in core flow resistance: when $D_{p,c}$ exceeds $D_{p,r}$, the core exhibits lower viscous resistance, increasing flow through the core region and driving more flow into the ring. Conversely, smaller $D_{p,c}/D_{p,r}$ ratios result in higher core viscous resistance, reducing f_c . This effect is consistent across all core sizes, as shown in the figure, with f_c rising as $D_{p,c}/D_{p,r}$ increases.

3.5. Model development

Building on the findings from the parametric study, a new model for flow distribution in dual-zone packed bed reactors was developed. The flow path of a fluid through a medium is governed by both viscous and inertial resistance. In a system with two regions of differing particle diameters, a stable state is reached when the backpressure in both regions is balanced, even though the flow distribution remains uneven,

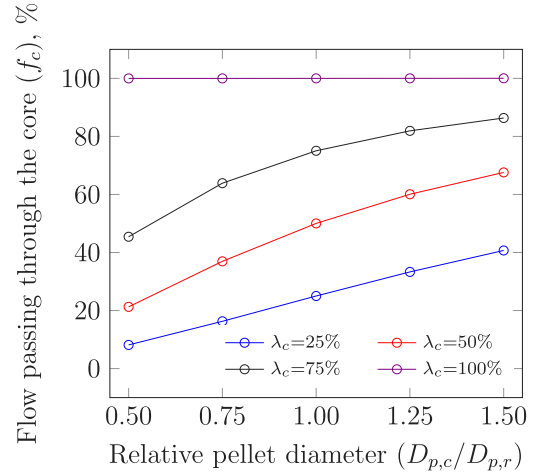


Fig. 10. Flow percent passing through the core in function of relative particle size with $Re_{in} = 100$.

as shown in Section 3.1. This backpressure equilibrium is expressed mathematically by Eq. (18).

$$\Delta p_c = \Delta p_r \quad (18)$$

Replacing both sides of the equation by Darcy–Forchheimer's law [47]:

$$\frac{\mu}{\alpha_c} u_c + C_c \frac{1}{2} \rho u_c^2 = \frac{\mu}{\alpha_r} u_r + C_r \frac{1}{2} \rho u_r^2 \quad (19)$$

Applying mass balance under the assumption of constant density, the inlet flow is equal to the sum of the flows through each region of the substrate:

$$A_T u_{in} = A_c u_c + A_r u_r \quad (20)$$

Solving for the ring velocity:

$$u_r = \left(\frac{A_T}{A_r} \right) u_{in} - \left(\frac{A_c}{A_r} \right) u_c \quad (21)$$

Defining the area fractions as $A_c/A_T = \lambda_c$ for the core region, and $A_r/A_T = (1 - \lambda_c)$ for the ring one, Eq. (21) can be written as:

$$u_r = \left(\frac{1}{1 - \lambda_c} \right) u_{in} - \left(\frac{\lambda_c}{1 - \lambda_c} \right) u_c \quad (22)$$

By substituting Eq. (22) into Eq. (19) and simplifying the expressions, the following model is obtained:

$$a u_c^2 + b u_c + c = 0 \quad (23)$$

where

$$a = \frac{C_c \rho}{2} - \frac{C_r \rho}{2} \left(\frac{\lambda_c}{1 - \lambda_c} \right)^2 \quad (24)$$

$$b = \frac{\mu}{\alpha_c} + \frac{\mu}{\alpha_r} \left(\frac{\lambda_c}{1 - \lambda_c} \right) + C_r \rho u_{in} \left(\frac{\lambda_c}{(1 - \lambda_c)^2} \right) \quad (25)$$

$$c = - \left[\frac{\mu u_{in}}{\alpha_r} \left(\frac{1}{1 - \lambda_c} \right) + \frac{C_r \rho u_{in}^2}{2} \left(\frac{1}{(1 - \lambda_c)^2} \right) \right] \quad (26)$$

Eq. (23) is solved using the quadratic formula, with only the positive root being physically meaningful. The negative solution would result in negative flow values or flows exceeding the total input, violating the mass balance.

$$u_c = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (27)$$

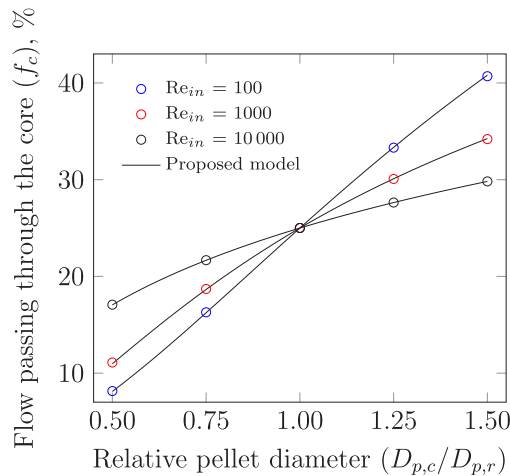


Fig. 11. Comparison of the proposed model and CFD data for several relative particle diameter and Re_{in} with $\lambda_c = 25\%$.

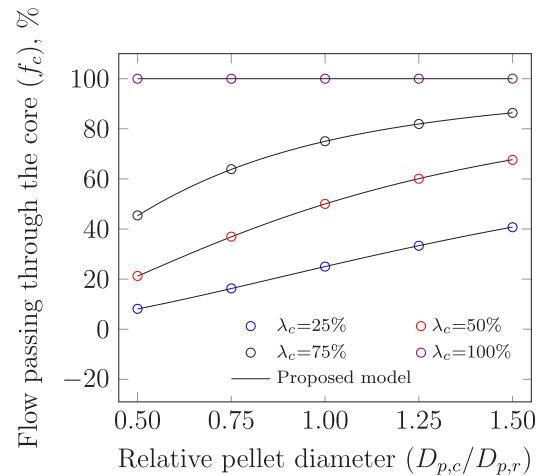


Fig. 12. Comparison of the proposed model and CFD data for several relative particle diameter and core sizes with $Re_{in} = 100$.

Additional solutions exist for specific cases, as outlined below:

if $a = 0$ ($\lambda_c = 0.5$ and $D_{p,c} = D_{p,r}$):

$$u_c = \frac{-c}{b} = u_{in} \quad (28)$$

if $\lambda_c \rightarrow 1$:

$$\lim_{\lambda_c \rightarrow 1} u_c = u_{in} \quad (29)$$

This solution was validated numerically for cases where the core size approached 1.

By substituting the velocity from Eqs. (23) to (26) into Eq. (17), the flow passing through the core is calculated and compared to CFD data. The results are presented in Figs. 11 and 12. The proposed model shows that all evaluated variables affect the flow distribution differently, depending on their combination, and the results are in excellent agreement with the CFD simulations.

In Fig. 11, the core size is fixed at 25%, and the flow through the core (f_c) is analyzed for several relative particle diameters ($D_{p,c}/D_{p,r}$) across different Reynolds numbers ($Re_{in} = 100, 1000$, and 10000). The model captures the non-linear trend in f_c with increasing particle diameter ratio, showing excellent agreement with the CFD data. Overall, the alignment between the model and CFD results demonstrates that the proposed model reliably predicts the effect of Reynolds number on flow distribution.

In Fig. 12, the Reynolds number is fixed at $Re_{in} = 100$, and the core flow is analyzed for varying particle size ratios ($D_{p,c}/D_{p,r}$) and different core sizes ($\lambda_c = 25\%, 50\%, 75\%$, and 100%). The proposed model again shows strong agreement with the CFD data, accurately capturing the increase in f_c with larger particle size ratios and core sizes. The model also predicts the convergence to 100% flow at $\lambda_c = 100\%$, representing a homogeneous bed, consistent with the CFD data. The agreement across all core sizes further validates the model's ability to generalize to different geometrical configurations.

A key observation is the importance of Re_{in} in the proposed model. This contrasts with the model developed for monoliths, where Re_{in} does not affect flow distribution [23]. Notably, if the values of C_c and C_r are set to zero in Eq. (23), the model reduces to that of a dual-cell monolith, demonstrating its versatility. Thus, the proposed model can be applied more broadly to any dual-zone porous medium with appropriate apparent permeability and inertial resistance coefficients.

4. Conclusions

The flow distribution in a non-homogeneous fixed bed with dual pellet sizes in the core and ring regions was successfully investigated using a computational model. The results indicate that the fraction of flow passing through the core is influenced by several factors, including core size, relative pellet diameter, and tube Reynolds number.

Larger core sizes result in higher flow percentages through the core, as the increased area reduces resistance compared to the ring region. At lower Reynolds numbers, the flow through the core is smaller, but it increases with higher Reynolds numbers, suggesting that higher flow rates promote more flow through the central region. This effect is more pronounced for larger core sizes, where the differences in flow percentage between Reynolds numbers are more significant.

The relative diameter of the pellets (core over ring) also plays a crucial role in flow distribution. When the relative diameter is less than 1.00, the flow through the core is reduced. As the relative diameter increases beyond 1.00, the flow through the core increases, although this effect diminishes at higher Reynolds numbers. A relative diameter of 1.00 leads to uniform flow distribution, where the percentage of flow through the core matches the core size.

A predictive model is developed to describe the flow distribution between the core and ring regions, accounting for both viscous and inertial resistances. The model is expressed as a quadratic equation, with coefficients dependent on the properties of the system, including core size, Reynolds number, pellet diameters, and fluid properties (density and viscosity).

Comparison of the proposed model with CFD simulation data shows excellent agreement across several core sizes, Reynolds numbers, and pellet diameters. This comparison suggests the model's potential for optimizing the design of packed beds with dual pellet sizes to achieve improved flow distribution. Future experimental validation is recommended to confirm the model's accuracy under a broader range of operating conditions in non-ideal scenarios. Since flow distribution dictates heat generation and transport in reactive systems, the extension of this work to include temperature distribution and conversion analysis is a logical next step. Future studies should also integrate these aspects to provide a comprehensive understanding of dual-zone reactor performance with significant thermal effects.

Nomenclature

a, b, c	Core velocity model constants
A	Cross-sectional area, m^2
A_c	Cross-sectional area of the core, m^2
A_r	Cross-sectional area of the ring, m^2
A_T	Total or tube cross-sectional area, m^2
C_i	Inertial resistance coefficients vector, m^{-1}
C_c	Core inertial resistance coefficient, m^{-1}
C_r	Ring inertial resistance coefficient, m^{-1}
D_p	Particle diameter, m
$D_{p,c}$	Core particle diameter, m
$D_{p,r}$	Ring particle diameter, m
f_c	Fraction of the flow passing through the core
L	Bed length, m
L_{in}	Inlet tube length, m
L_{out}	Outlet tube length, m
L_s	Porous medium tube length, m
p	Pressure, Pa
Δp_c	Core pressure drop, Pa
Δp_r	Ring pressure drop, Pa
R_c	Core radius, m
R_t	Total or tube radius, m
Re_{in}	Inlet or tube Reynolds number, $= \rho u_{in}(2R_t)/\mu$
Re_p	Particle Reynolds number, $= \rho u_{in}D_p/\mu$
u	Velocity magnitude, m/s
u_i	Velocity vector, m/s
u_{in}	Inlet superficial velocity, m/s
u_c	Core superficial velocity, m/s
u_r	Ring superficial velocity, m/s
α_i	Apparent permeability vector, m^2
α_c	Core apparent permeability, m^2
α_r	Ring apparent permeability, m^2
ε	Bed porosity or void fraction
λ_c	Core fractional size
Φ	Bed sphericity
ρ	Density, kg/m^3
μ	Viscosity, Pa·s

CRediT authorship contribution statement

Pablo Díaz: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Consuelo Reinao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Iván Cornejo:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Acknowledgments

The author Pablo Díaz is grateful for the financial support provided by DPP UTFSM through Programa de Incentivo a la Iniciación Científica (PIIC).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] S. Arab, J.-M. Commenge, J.-F. Portha, L. Falk, Methanol synthesis from CO₂ and H₂ in multi-tubular fixed-bed reactor and multi-tubular reactor filled with monoliths, *Chem. Eng. Res. Des.* 92 (11) (2014) 2598–2608.
- [2] D. Iranshahi, A. Golrokh, E. Pourazadi, S. Saeidi, F. Gallucci, Progress in spherical packed-bed reactors: Opportunities for refineries and chemical industries, *Chem. Eng. Processing- Process. Intensif.* 132 (2018) 16–24.
- [3] W. Chansiriwat, K. Wantala, R. Khunphonoi, P. Khemthong, C. Lorentz, B. da Costa Magalhães, M. Prevot, D. Laurenti, C. Geantet, Gypsum-derived CaO catalytic pyrolysis of palm oil in a continuously packed bed reactor, *J. Anal. Appl. Pyrolysis* 178 (2024) 106414.
- [4] M. Mokashi, A.B. Shirsath, A. Çelik, P. Lott, H. Müller, S. Tischer, L. Maier, J. Bode, D. Schlereth, F. Scheiff, et al., Methane pyrolysis in packed bed reactors: Kinetic modeling, numerical simulations, and experimental insights, *Chem. Eng. J.* 485 (2024) 149684.
- [5] S.B. Jaffe, Hot spot simulation in commercial hydrogenation processes, *Ind. Eng. Chem. Process. Des. Dev.* 15 (3) (1976) 410–416.
- [6] M. Shah, D. West, V. Balakotaiah, Analysis of thermoflow instabilities in adiabatic packed-bed catalytic reactors, *Chem. Eng. J.* 452 (2023) 139365.
- [7] P. Forzatti, L. Lietti, Catalyst deactivation, *Catal. Today* 52 (2–3) (1999) 165–181.
- [8] C.H. Bartholomew, Mechanisms of catalyst deactivation, *Appl. Catal. A: Gen.* 212 (1–2) (2001) 17–60.
- [9] G.A. Tafete, N.G. Habtu, Reactor configuration, operations and structural catalyst design in process intensification of catalytic reactors: A review, *Chem. Eng. Processing- Process. Intensif.* 184 (2023) 109290.
- [10] R. Agrawal, D.H. West, V. Balakotaiah, Modeling and analysis of local hot spot formation in down-flow adiabatic packed-bed reactors, *Chem. Eng. Sci.* 62 (18–20) (2007) 4926–4943.
- [11] R. Henda, A. Machac, B. Nilsson, Heat and mass transport in a nonlinear fixed-bed catalytic reactor: Hot spots and thermal runaway, *Chem. Eng. J.* 143 (1–3) (2008) 195–200.
- [12] M. Ambrosetti, M. Bracconi, M. Maestri, G. Groppi, E. Tronconi, Packed foams for the intensification of catalytic processes: assessment of packing efficiency and pressure drop using a combined experimental and numerical approach, *Chem. Eng. J.* 382 (2020) 122801.
- [13] M. Ambrosetti, G. Groppi, W. Schwieger, E. Tronconi, H. Freund, Packed periodic open cellular structures—an option for the intensification of non-adiabatic catalytic processes, *Chem. Eng. Processing- Process. Intensif.* 155 (2020) 108057.
- [14] R. Balzarotti, M. Ambrosetti, A. Beretta, G. Groppi, E. Tronconi, Investigation of packed conductive foams as a novel reactor configuration for methane steam reforming, *Chem. Eng. J.* 391 (2020) 123494.
- [15] A. Aguirre, V. Chandra, E. Peters, J. Kuipers, M.N. D'Angelo, Open-cell foams as catalysts support: A systematic analysis of the mass transfer limitations, *Chem. Eng. J.* 393 (2020) 124656.
- [16] C. Ferroni, M. Bracconi, M. Ambrosetti, M. Maestri, G. Groppi, E. Tronconi, A fundamental investigation of gas/solid heat and mass transfer in structured catalysts based on periodic open cellular structures (POCS), *Ind. Eng. Chem. Res.* 60 (29) (2021) 10522–10538.
- [17] C.W.M. Quintero, G. Ercolino, A. Poozhikunnath, R. Maric, S. Specchia, Analysis of heat and mass transfer limitations for the combustion of methane emissions on PdO/Co₃O₄ coated on ceramic open cell foams, *Chem. Eng. J.* 405 (2021) 126970.
- [18] A. Hadi, R. Emadi, S. Baghshahi, S. Naghavi, Different pore size alumina foams and study of their mechanical properties, *Ceramics- Silik.* 59 (1) (2015) 6–9.
- [19] K. Dammler, K. Schelm, U. Betke, T. Fey, M. Scheffler, Open-cellular alumina foams with hierarchical strut porosity by ice templating: A thickening agent study, *Materials* 14 (2021) 1060.
- [20] R. Yeetsorn, S. Tungkamani, Y. Maiket, Fabrication of a ceramic foam catalyst using polymer foam scrap via the replica technique for dry reforming, *ACS Omega* 7 (5) (2022) 4202–4213.
- [21] W. Zhang, Y. Lin, Y. Zhang, T. Li, J. Li, Z. Chen, K. Norinaga, Regulation of temperature distribution in fixed bed reactor for CO₂ methanation through “CHES” monolith structure catalyst, *Appl. Therm. Eng.* 236 (2024) 121826.
- [22] I. Cornejo, G. Garreton, R.E. Hayes, On the use of dual cell density monoliths, *Catalysts* 11 (9) (2021) 1075.
- [23] C. Reinao, I. Cornejo, A model for the flow distribution in dual cell density monoliths, *Processes* 11 (3) (2023) 827.
- [24] R. Beetstra, M.A. van der Hoef, J. Kuipers, Numerical study of segregation using a new drag force correlation for polydisperse systems derived from lattice-Boltzmann simulations, *Chem. Eng. Sci.* 62 (1–2) (2007) 246–255.
- [25] S. Gao, J. Theuerkauf, P. Pakseresh, K. Kellogg, Y. Fan, A modified Ergun equation for application in packed beds with bidisperse and polydisperse spherical particles, *Powder Technol.* 445 (2024) 120035.
- [26] S. Ergun, Fluid flow through packed columns, *Chem. Eng. Prog.* (1952) 89.
- [27] E. Erdim, Ö. Akgiray, İ. Demir, A revisit of pressure drop-flow rate correlations for packed beds of spheres, *Powder Technol.* 283 (2015) 488–504.
- [28] I. Macdonald, M. El-Sayed, K. Mow, F. Dullien, Flow through porous media—the Ergun equation revisited, *Ind. Eng. Chem. Fundam.* 18 (3) (1979) 199–208.

- [29] NSSC KTA, Reactor core design of high-temperature gas-cooled reactors part 3: Loss of pressure through friction in pebble bed cores, 1981, Germany.
- [30] E. Ozahi, M.Y. Gundogdu, M.Ö. Carpinlioglu, A modification on Ergun's correlation for use in cylindrical packed beds with non-spherical particles, *Adv. Powder Technol.* 19 (4) (2008) 369–381.
- [31] H. Rose, On the resistance coefficient–Reynolds number relationship for fluid flow through a bed of granular material, *Proc. Inst. Mech. Eng.* 153 (1) (1945) 154–168.
- [32] L. Gibilaro, R. Di Felice, S. Waldram, P.U. Foscolo, Generalized friction factor and drag coefficient correlations for fluid-particle interactions, *Chem. Eng. Sci.* 40 (10) (1985) 1817–1823.
- [33] A. Montillet, Flow through a finite packed bed of spheres: a note on the limit of applicability of the Forchheimer-type equation, *J. Fluids Eng.* 126 (1) (2004) 139–143.
- [34] L.D. Harrison, K.M. Brunner, W.C. Hecker, A combined packed-bed friction factor equation: Extension to higher Reynolds number with wall effects, *AIChE J.* 59 (3) (2013).
- [35] K. Mohammadpour, A. Alkhalaf, E. Specht, CFD simulation of cross-flow mixing in a packed bed using porous media model and experimental validation, *Comput. Part. Mech.* 6 (2019) 157–162.
- [36] W.L. McCabe, J.C. Smith, P. Harriott, *Unit Operations of Chemical Engineering*, McGraw-Hill, New York, 2005.
- [37] S. Fleischlen, T. Turek, G.D. Wehinger, Local structure effects on hydrodynamics in slender fixed bed reactors: Spheres and rings, *Chem. Eng. J.* 475 (2023) 146342.
- [38] J. Gorman, A. Zheng, E. Sparrow, Bounding wall effects on fluid flow and pressure drop through packed beds of spheres, *Chem. Eng. J.* 373 (2019) 519–530.
- [39] S. Afandizadeh, E. Foumeny, Design of packed bed reactors: guides to catalyst shape, size, and loading selection, *Appl. Therm. Eng.* 21 (6) (2001) 669–682.
- [40] J. von Seckendorff, O. Hinrichsen, Review on the structure of random packed-beds, *Can. J. Chem. Eng.* 99 (2021) S703–S733.
- [41] C. Wu, Y. Ferng, C. Chieng, C. Liu, Investigating the advantages and disadvantages of realistic approach and porous approach for closely packed pebbles in CFD simulation, *Nucl. Eng. Des.* 240 (5) (2010) 1151–1159.
- [42] ANSYS Inc., ANSYS Fluent User's Guide Release 2023R2, Canonsburg, PA, USA, 2023.
- [43] F.R. Menter, Improved Two-Equation K-Omega Turbulence Models for Aerodynamic Flows, *Tech. Rep.*, 1992.
- [44] F.R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, *AIAA J.* 32 (8) (1994) 1598–1605.
- [45] N. Jurtz, M. Kraume, G.D. Wehinger, Advances in fixed-bed reactor modeling using particle-resolved computational fluid dynamics (CFD), *Rev. Chem. Eng.* 35 (2) (2019) 139–190.
- [46] A.G. Dixon, B. Partopour, Computational fluid dynamics for fixed bed reactor design, *Annu. Rev. Chem. Biomol. Eng.* 11 (1) (2020) 109–130.
- [47] F.M. White, *Fluid Mechanics*, McGraw-Hill, New York, NY, USA, 2009.
- [48] E.M. Moghaddam, E. Foumeny, A. Stankiewicz, J. Padding, Fixed bed reactors of non-spherical pellets: importance of heterogeneities and inadequacy of azimuthal averaging, *Chem. Eng. Sci.: X* 1 (2019) 100006.